

What Is a Graph?

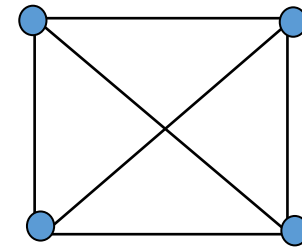
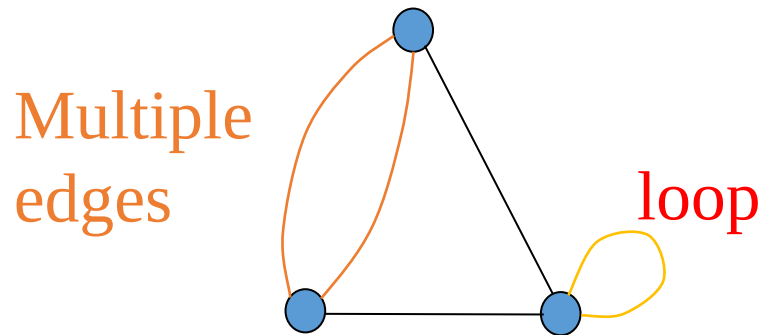
- A graph is a generic model for solving discrete problems
 - A set of vertices
 - A set of edges
- A **vertex** : *an object*
- An **edge** : *a binary relation (between objects)*

What Is a Graph?

- A more formal definition is as follows:
- A graph G is a triplet consisting of:
 - A vertex set $V(G)$
 - An edge set $E(G)$, **disjoint from the vertex set**
 - A relation between an edge and a pair of vertices (not necessarily distinct); these vertices are referred to as the endpoints or end-vertices of the edge
 - General notation : $|V(G)| = n$, $|E(G)| = m$

Loop, Multiple edges

- *Loop* : An edge whose endpoints are equal
- *Multiple edges* : Edges have the same pair of endpoints
- *Simple Graph* : No loops, No Multiple edges



Finite Graph, Null Graph

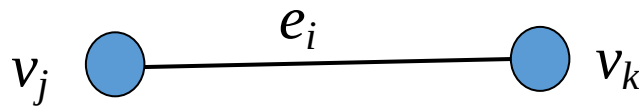
- *Finite graph* : a graph whose vertex set and edge set are finite
- *Null graph* : a graph whose vertex set and edge set are empty
- We will assume (unless explicitly stated to the contrary) that all graphs are **finite**, and that the vertex set is ***non-empty***.
However, we will allow the edge set to be empty.

Simple Graphs

- **Remark:** We have introduced the notion of a **graph** G as a set of vertices $V(G)$, a set of edges $E(G)$, and an assignment or function from $E(G)$ to 1-element or 2-element subsets of $V(G)$. Note that in the textbook M&N, it is assumed that all graphs are **simple graphs**, in which the subsets have to be 2-element subsets (*no loops*), and the assignment function has to be injective (*no multiple edges*). In this case, the edge set is a subset of the two-element subsets of the vertex set $V(G)$. In other words, $E(G) \subseteq B(V(G), 2)$ in terms of our previously introduced notation. We will not follow the usage of M&N, and will permit both loops and multiple edges. However, some of the concepts and results are meaningful for simple graphs only, and this will be stated whenever applicable.

Adjacency, Incidence, and Degree

- Assume e_i is an edge whose endpoints are v_j, v_k
- The vertices v_j and v_k are said to be **adjacent**
- The edge e_i is said to be **incident upon** v_j (and also v_k)
- **Degree** of a vertex v_k is the number of edges incident upon v_k . It is denoted as $d(v_k)$ – usually used only for loop-less graphs.



Basic Result about Vertex Degrees

Proposition 30: Let G be a (loop-less) graph. Then, the sum of the degrees of the vertices is twice the number of edges, i.e.

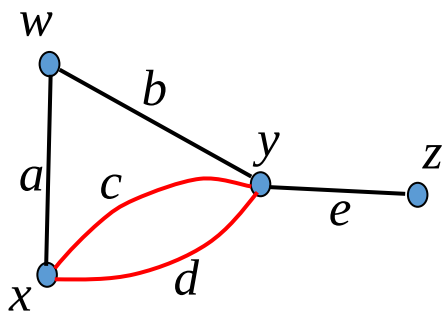
$$\sum_{v \in V(G)} d(v) = 2 |E(G)|$$

Proof: Each edge contributes 2 degrees (to the vertices of G).

Remark: We can apply the above result to graphs with loops as well, provided we adopt the convention that a loop contributes two degrees to its endpoint (NB: ***In a loop, the two endpoints are the same vertex.***

Adjacency Matrix

- Let $G = (V, E)$, $|V| = n$ and $|E| = m$
- The **adjacency matrix** of G written $A(G)$, is the n -by- n matrix in which entry $a_{i,j}$ is the number of edges in G with endpoints $\{v_i, v_j\}$.



$$\begin{matrix} & w & x & y & z \\ \begin{matrix} w \\ x \\ y \\ z \end{matrix} & \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & \color{red}{2} & 0 \\ 1 & \color{red}{2} & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix}$$

G a simple graph
(no loops, no multiple edges)

Then: the adjacency matrix $A(G)$
is a symmetric $(0, 1)$ -matrix.

→ Symmetric Matrices — An
important Result → (for real matrices)
Orthogonally Diagonalizable over
 $\mathbb{R}$

Complement

- **Complement of G :** The complement G' of a simple graph G :

- A simple graph
- $V(G') = V(G)$
- $E(G') = \{ uv \mid uv \notin E(G) \}$

Every edge is determined by its endpoints

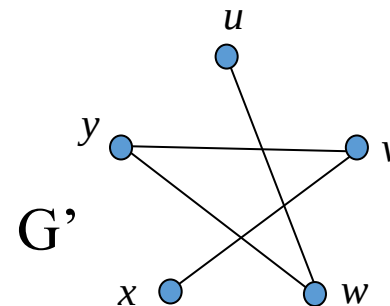
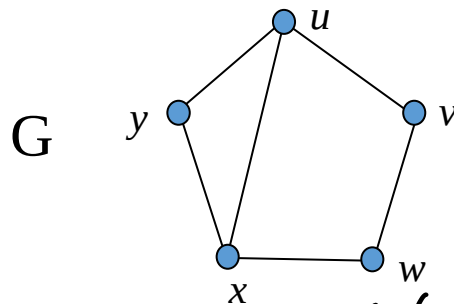
$$e = uv$$

If $|V(G)| = n$, then

$$m = |E(G)|$$

$$\leq \binom{n}{2}$$

n choose 2



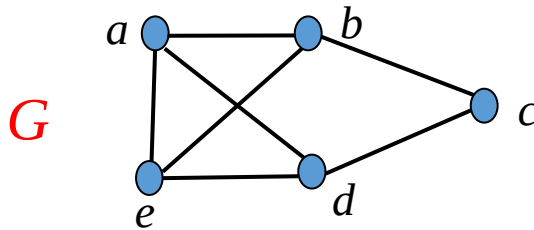
$$\text{Then } |E(G')| = \binom{n}{2} - m = n(n-1)/2$$

Subgraphs - 1

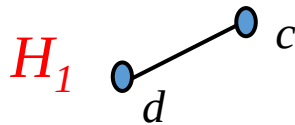
- A **subgraph** of a graph G is a graph H such that:
 - $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$ and
 - The assignment of endpoints to edges in H is the same as in G
 - *In the case of simple graphs, each edge is identified with its two distinct endpoints. Hence, the second bullet above is not really needed*
- An **induced subgraph** of G is a subgraph H of G such that $E(H)$ consists of all edges of G whose endpoints belong to $V(H)$

Subgraphs - 2

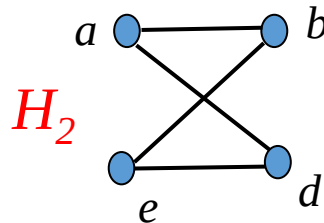
- Example: H_1 , H_2 , and H_3 are subgraphs of G



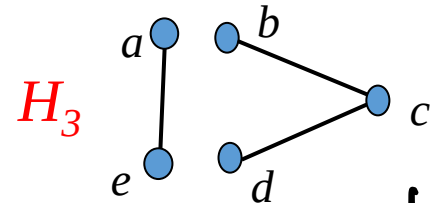
Is any of these an induced subgraph?



YES



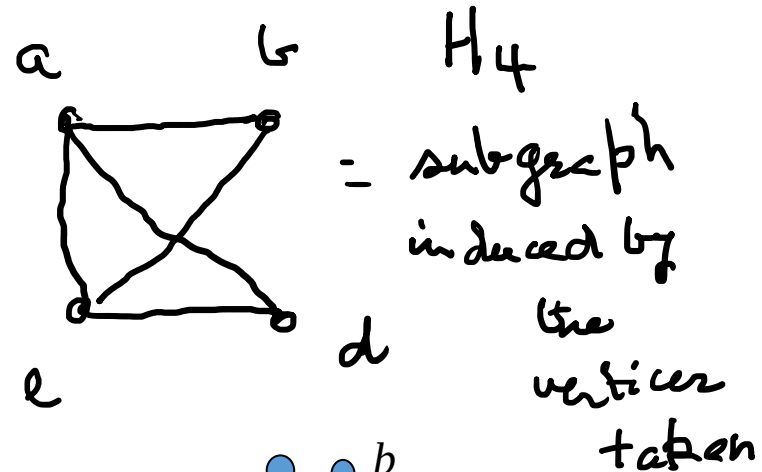
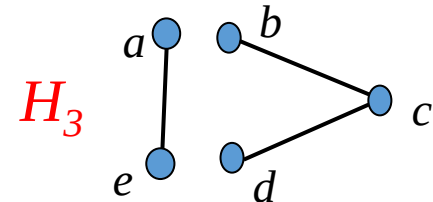
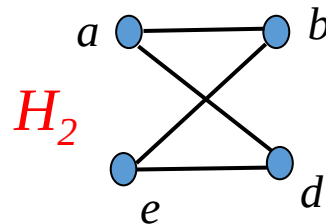
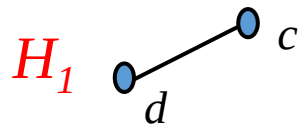
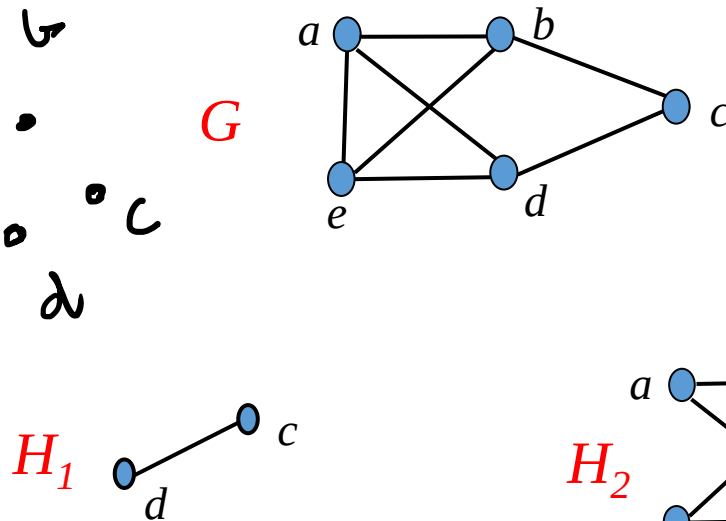
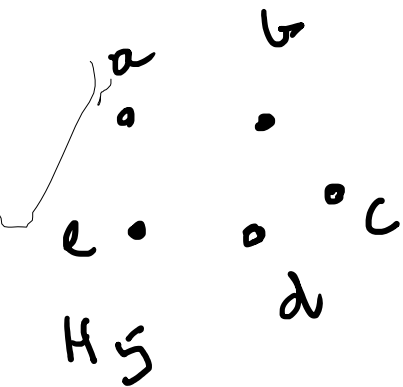
NO



NO!

Subgraphs - 3

- Example: H_1 , H_2 , and H_3 are subgraphs of G

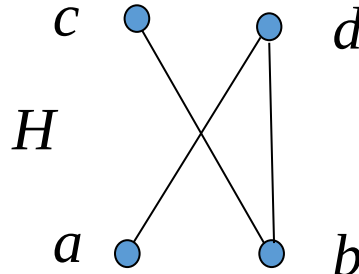
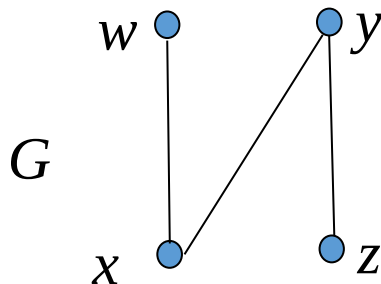


Proper subgraph -
other than G

$e(H_2)$ does not include all possible edges of G w/ these points

Isomorphism - 1

- An **isomorphism** from a simple graph G to a simple graph H is a bijection $f: V(G) \rightarrow V(H)$ such that $uv \in E(G)$ if and only if $f(u)f(v) \in E(H)$
 - We say “ **G is isomorphic to H** ”, written $G \cong H$
 - Is isomorphism an equivalence relation?



$f_1: \begin{matrix} w & x & y & z \\ c & b & d & a \end{matrix}$

$f_2: \begin{matrix} w & x & y & z \\ a & d & b & c \end{matrix}$

An example ~~for two graphs G and H~~

$$G \rightarrow V(G) = \{u_1, u_2, u_3, u_4\}$$

$$E(G) = \{u_1 u_2, u_1 u_3, u_1 u_4, u_2 u_3, u_2 u_4\}$$

$$H \rightarrow V(H) = \{v_1, v_2, v_3, v_4\}$$

$$E(H) = \{v_1 v_2, v_1 v_3, v_1 v_4, v_2 v_3, v_3 v_4\}$$

$$A(G) = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \quad A(H) = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$
$$A(G) \neq A(H)$$

Apply a simultaneous permutation to
rows and columns of $A(H)$

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} R_2 \leftrightarrow R_3 \\ C_2 \leftrightarrow C_3 \end{matrix} \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} = A(G)$$

Hence, $G \cong H$

Looks easy - but in fact difficult!
- this provides another way to check isomorphism

Isomorphism - 2

- **YES !** Isomorphism is an equivalence relation on the set of all graphs with the same number of vertices.
- An ***isomorphism class*** of graphs is an equivalence class under the isomorphism relation.
- When we draw a graph, we assign names to the vertices (and hence the edges). Such a graph is referred to as a **labeled** graph. However, its structural properties also apply to every isomorphic graph. Hence we use the informal expression **unlabeled** graph to refer to an isomorphism class.
- This allows us to refer conveniently to various classes of graphs, e.g. K_n – the complete graphs with n vertices, P_n – the paths with n vertices, C_n – the cycles with n vertices, $K_{r,s}$ – the complete bipartite graphs with classes or partite sets of order r,s , etc.

Isomorphism - 3

- It is easy to determine the number of distinct labeled simple graphs with n vertices. In this case, an edge corresponds to a subset with 2 vertices. If there are n vertices, then the number of such 2-subsets is the binomial coefficient “ n choose 2”, written $B(n,2)$
- Since we can either choose or not choose a particular edge, the number of labeled graphs is 2^k , where $k = B(n,2)$
- Finding how many of these are isomorphic to each other is much harder. It is not known whether the graph isomorphism problem can be solved in polynomial time or not, nor is it known whether the graph isomorphism problem is NP-complete. However, the so-called subgraph isomorphism problem, i.e. whether a graph contains a subgraph isomorphic to a given graph is known to be NP-complete.
- However, for certain special classes of graphs - such as trees and planar graphs, which we will not be able to study in this course - there are known polynomial time algorithms.

Isomorphism - 4

- We can solve the problem for graphs with a small number of vertices by brute force.
- In the notes, we will present a list of the 11 isomorphism classes of graphs with 4 vertices, whereas there are 64 labeled graphs with 4 vertices.
- For the above, we use the fact that $G \cong H$ if and only if $G' \cong H'$. (**Left as an exercise**).
- We will also present a list of some well-known isomorphism classes of graphs with a small number of vertices (these occur in problems).

Automorphism

- An *automorphism* of G is a permutation of $V(G)$ that is an isomorphism from G to G .
 - A graph is called *vertex transitive* if for every pair $u, v \in V(G)$ there is an automorphism that maps u to v .
 - *For example, the complete bipartite graph $K_{n,n}$ is vertex transitive. A more interesting example will be presented in the tutorial problems. P_3 is a nice and easy example of a graph which is not vertex transitive.*